AI Radiology Triage System

A clinical workflow platform that helps radiologists identify potentially urgent chest X-rays earlier while keeping interpretation and care decisions with the radiologist.

1. Problem

Radiology teams manage high volumes of chest X-rays. Potentially urgent studies can wait behind routine cases when all examinations enter a standard queue. Orders, images, AI findings and radiologist reviews may also pass through separate workflow stages, making it harder to track each case from acquisition to final report.

Urgent studies in routine queues: A standard queue may not make potentially urgent examinations visible early enough for review.

Fragmented workflow: Imaging orders, DICOM images, screening results and radiologist reviews can be spread across different systems or stages.

Limited visual guidance: Radiologists may need to locate candidate suspicious regions themselves while reviewing each image.

Coordination burden: Teams need a traceable link between the order, image, prioritisation result and final interpretation.

2. Solution

The system screens supported chest X-rays, assigns a priority level and highlights candidate suspicious regions to help radiologists decide which studies to review sooner. It brings the order, DICOM image, AI result, worklist status and final review into one traceable workflow.

The system is designed to work alongside existing diagnostics, RIS/PACS and clinical systems. It supports PA, AP and PA + AP chest X-ray workflows. Clinical responsibility remains with the radiologist: the AI supports prioritisation and does not make the diagnosis.

3. Platform Features

The platform combines image screening, worklist prioritisation and review tools in one radiology workflow.

Feature	What it offers
Chest X-ray screening	Screens supported PA, AP and PA + AP studies using the configured AI pipeline.
Priority assignment	Assigns a priority level so studies that may need earlier attention can be surfaced in the worklist.
Candidate-region highlighting	Marks candidate suspicious regions on the image to guide the radiologist's review.
Radiologist worklist	Organises studies by priority to support review order and team coordination.
Traceable case workflow	Links the imaging order, DICOM image, AI output, radiologist review and final report.
Clinical oversight	Keeps final image interpretation and reporting with the radiologist.

Together, these features make potentially urgent studies easier to identify and follow through the review process.

Area	Capability	Expected benefit
Image analysis	DenseNet121 screening and YOLO11n candidate-region detection	Provides a priority signal and visual cues for radiologist review.
Image workflow	DICOM support with Orthanc storage and imaging services	Keeps medical images available within the connected workflow.
Image review	OHIF medical image viewer	Allows radiologists to inspect studies and AI highlights.
Application	React.js interface with FastAPI / Python workflow services	Presents worklists and coordinates screening and review steps.
System integration	Designed to work alongside diagnostics, RIS/PACS and clinical systems	Supports coordination with existing care and imaging operations.

4. Intended Impact

The system is intended to support faster attention to potentially urgent studies, improve worklist prioritisation, reduce the chance that important cases remain in routine queues, and improve coordination among doctors, diagnostics teams and radiologists. By connecting each order, image, AI result and final review, it also supports end-to-end traceability and more focused use of specialist time.

5. How the System Works

A chest X-ray moves through the following workflow from order to final report:

Stage	Activity	Workflow output
1. X-ray order	A chest X-ray is requested in the clinical workflow.	An order is available for imaging.
2. Image acquisition	The supported chest X-ray view or views are acquired (PA, AP or PA + AP).	A study is created.
3. DICOM storage	The image is stored and made available through the imaging service.	The DICOM study is ready for screening and review.
4. AI screening	The configured screening and candidate-region models process the study.	A priority level and candidate-region output are returned.
5. Priority assignment	The study is placed in the radiologist worklist according to priority.	The worklist supports review sequencing.
6. Radiologist review	The radiologist opens the image, considers the AI output and performs the clinical interpretation.	The radiologist determines the findings.
7. Final report	The final interpretation is recorded and linked to the study workflow.	The order-to-report path remains traceable.

AI prioritises. The radiologist decides.

6. Technology Foundation and Integration

The platform is described as a custom AI-enabled radiology application. Its user interface is built with React.js, while FastAPI and Python provide backend and workflow services. PostgreSQL is the database. DenseNet121 supports chest X-ray screening, and YOLO11n identifies candidate regions. Orthanc provides DICOM image storage and imaging services, and OHIF provides medical image viewing.

DICOM is used for medical imaging exchange, and the solution is designed to work alongside existing diagnostics, RIS/PACS and clinical systems.

7. Model Details and ROC-AUC

The system uses an image classification model and an object detection model for different tasks. Their outputs are supporting signals for the radiologist's review.

Model	Task	Description	Output in the workflow
DenseNet121	Image classification	A 121-layer convolutional neural network. Dense connections pass feature maps between layers so later layers can reuse earlier image features. In this system it screens the complete chest X-ray.	Image-level class prediction or score used by the screening workflow.
YOLO11n	Object detection	The nano-sized YOLO11 model variant configured to detect and locate candidate regions in an image. It processes the radiograph for localized visual findings.	Candidate bounding boxes with predicted class and confidence, displayed as review aids.

A detection box marks a model-identified candidate region; it does not confirm a clinical finding. The radiologist reviews the full study and decides whether the highlighted region is relevant.

What ROC-AUC means

The receiver operating characteristic (ROC) curve plots the true positive rate (sensitivity) against the false positive rate (1 minus specificity) as the decision threshold changes. The area under that curve, or ROC-AUC, summarizes how well a model ranks positive cases above negative cases across thresholds. A value of 0.5 represents chance-level ranking, while 1.0 represents perfect separation on the evaluated data.

Measure	Definition	Why it matters for triage
Sensitivity	True positives divided by all actual positives.	Shows how many urgent cases the chosen threshold detects.
Specificity	True negatives divided by all actual negatives.	Shows how many non-urgent cases remain unflagged.
ROC-AUC	Area under the sensitivity-versus-false-positive-rate curve across thresholds.	Summarizes ranking performance; it does not select the operating threshold.

ROC-AUC is not the same as accuracy, a confidence calibration measure or a clinical outcome. For a meaningful report, calculate it on an independent, representative test set and state the target label and evaluation method. Also report sensitivity, specificity and workload at the threshold selected for the clinical workflow. A model name alone does not establish a numerical ROC-AUC result.

Technical References

DenseNet: Huang et al., CVPR 2017, Densely Connected Convolutional Networks, https://openaccess.thecvf.com/content_cvpr_2017/html/Huang_Densely_Connected_Convolutional_CVPR_2017_paper.html
YOLO11: Ultralytics model documentation, https://docs.ultralytics.com/models/yolo11/
ROC-AUC: scikit-learn metric documentation, https://scikit-learn.org/stable/modules/generated/sklearn.metrics.roc_auc_score.html